

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

Roll No. : S031	Name: Panvir
Class : SYIT	Batch : 01
Date of Assignment : 12-8-26	Date/Time of Submission : 12-8-26

Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object. \

Code:-

```
import pandas as pd

data = {
    "name": ["Sally", "Mary", "John"],
    "age": [50, 40, 30],
    "qualified": [True, False, False]
}
df = pd.DataFrame(data)
print(df)
print("Panvir S031")
```

Output:-

```
   name  age  qualified
0  Sally   50      True
1  Mary   40     False
2  John   30     False
Panvir S031
|
```

5b. Write a python code to show statistical information on given data set.

Code:-

```
import statistics
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.mode([1, 3, 3, 3, 5, 7, 9, 11]))
print(statistics.stdev([1, 3, 5, 7, 9, 11]))
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print("Panvir S031")
```

Output:-

```
7
7
3
3.7416573867739413
14
Panvir S031
|
```

5c. Write a python code to create pandas series from dictionaries.

Code:-

```
import pandas as pd

dictionary = {'A': 10, 'B': 20, 'C': 30}

series = pd.Series(dictionary)

print(series)
print("Panvir S031")
```

Output:-

```
A      10
B      20
C      30
dtype: int64
Panvir S031
|
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

Code:-

```
import pandas as pd

dict = {'name': ["Panvir", "Ayush", "Soham", "Viraj"],
        'degree': ["Bscit", "BCA", "M.Tech", "MBA"],
        'score': [90, 40, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df)
print("Panvir S031")
```

Output:-

```
name degree score
True  Panvir   Bscit    90
False Ayush    BCA    40
True   Soham  M.Tech    80
False Viraj    MBA    98
Panvir S031
```

